BTS (C) – VI – 10 – 008 – P

B. Tech Degree VI Semester Examination, April 2010

ME 604 HEAT AND MASS TRANSFER

(2006 Scheme)

Time : 3 Hours Maximum Marks : 100

(Use of Heat and Mass Transfer Data Book permitted)

PART - A

(Answer ***ALL*** questions)

(8 x 5 = 40)

I. (a) Derive an expression for critical radius of a sphere.
(b) Explain about lumped heat capacity system.
(c) Explain about thermal and hydrodynamic boundary layers.
(d) Differentiate between film wise and drop wise condensation.
(e) State and prove Stefan-Boltzman law.
(f) Explain about view factor.
(g) Explain about fouling factor in a heat exchanger analysis.
(h) What do you mean by effectiveness of a heat exchanger?

PART – B

(4 x 15 = 60)

II. (a) A composite wall has 25 cm of brick and 5 cm of concrete. The brick side is exposed to 50°C and concrete side is exposed to 5° C. Determine the heat loss through 50 m^2 area of the wall and the interface temperature. Assume K for brick as .7 W/mK and concrete as .93 W/mK. (5)
(b) Derive the heat conduction equation in cylindrical co-ordinate system. (10)

OR

III. (a) Explain about the physical significance of Biot number and Fourier number. (5)
(b) Derive an expression for temperature distribution and heat transfer rate for a infinitely long fin. (10)

IV. (a) Explain about the physical significance of Reynold's number, Prandtl number and Grashof number. (5)
(b) Air at a pressure of 6 kN/m^2 and a temperature of 300°C flows with a velocity of 10 m/sec over a flat plate .5 m long. Estimate the cooling rate per unit width of the plate needed to maintain it at a surface temperature of 27°C. (10)

OR

V. (a) Explain Reynolds analogy for turbulent heat transfer. (5)
(b) A vertical plate of 60 cm height is at 160°C and is exposed to still air, at 100°C. Determine heat transfer rate per unit width. (10)

VI. (a) Explain about heat transfer in the presence of re-radiating surfaces. (5)
(b) Derive the relationship for shape factor namely

$$Fij = \frac{1}{A_i} \int_{Ai}\int_{Aj} \frac{\cos\theta_i \cos\theta_j}{\pi R^2} \, dA_i dA_j$$ (10)

OR

VII. (a) State and prove reciprocity theorem. (5)
(b) A furnace cavity which is in the form of a cylinder of 75 mm diameter and 150 mm length is open at one end to large surroundings at 27°C. The sides and bottom are approximated as black bodies and are heated electrically, well insulated and maintained at temperatures of 1350 and 1650 respectively. How much power is required to maintain the furnace conditions? (10)

VIII. (a) Derive an expression for L.M.T.D. of a counter flow heat exchanger. (8)
(b) Derive an expression for effectiveness of a parallel flow heat exchanger. (7)

OR

IX. A finned tube cross flow heat exchanger has gas side overall heat transfer coefficient and area as 100 w/m^2K and 40 m^2. The water flow rate and inlet temperature are 1 Kg/sec and 35°C. Hot gases enter heat exchanger at 1.5 Kg/sec and at a temperature of 250°C. The specific heat of gas is 1000 J/Kg°K. Determine the heat transfer rate and outlet temperatures. (15)

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